

# Project Title:

# Linux Host Security Investigation

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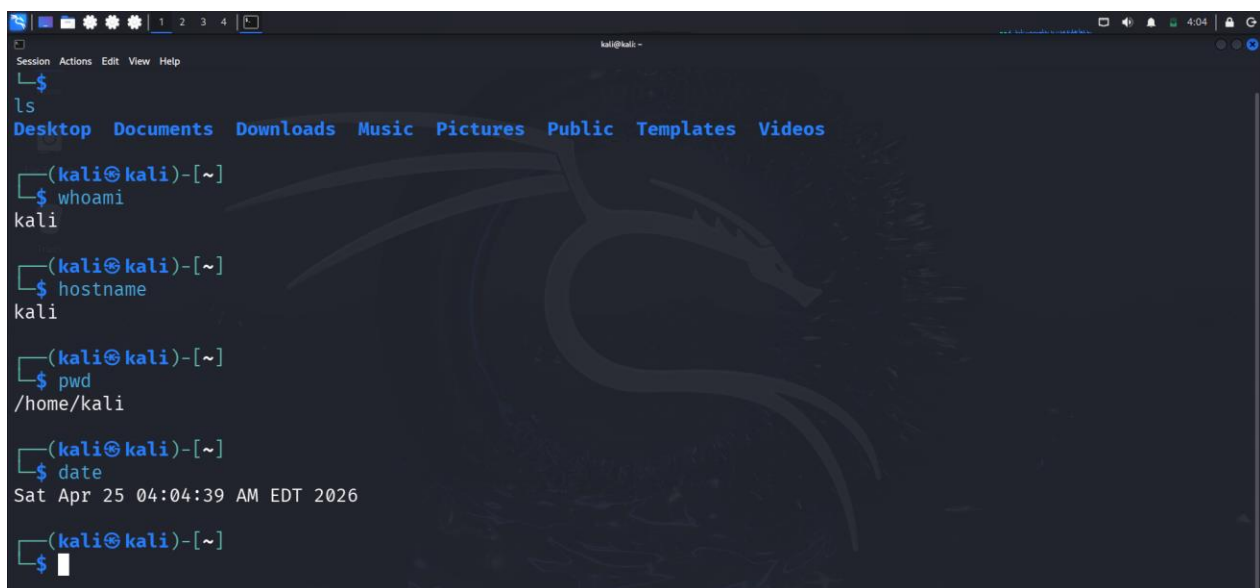
*Investigator Name: Syed Qammar Abbas*

*Date: 25 April , 2026*

*System Name: kali-VirtualBox*

*Operating System:kali linux(debian)*

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```
kali@kali -
Session Actions Edit View Help
└─$
ls
Desktop Documents Downloads Music Pictures Public Templates Videos
└─(kali@kali)-[~]
└─$ whoami
kali
└─(kali@kali)-[~]
└─$ hostname
kali
└─(kali@kali)-[~]
└─$ pwd
/home/kali
└─(kali@kali)-[~]
└─$ date
Sat Apr 25 04:04:39 AM EDT 2026
└─(kali@kali)-[~]
└─$
```

## OBJECTIVE:

The purpose of this project is to practice the investigation of a linux system and practice the file system , commands , log reading and investigation and checking the security by permission and process monitoring .

## CREATING SUSPICIOUS ENVIRONMENT:

We did :

- ⑨ Authentication Failure Simulation
- ⑨ Weak Permission File(Insecure.txt)
- ⑨ Background Suspicious Process (PID = 6393)
- ⑨ Fake Sudo log (sudo ls /root)

## FINDINGS:

So , Current Findings are Found by investigating the system :-

### Finding 1: File System Investigation

*Command Used:*

- I) *ls (List of all the visible files and directories )*
- II) *ls -l (To See the permissions as well as owner)*
- III) *ls -la (For hidden Directories and files ) Evidence:*

*The system had no strange or weird files .*

*Analysis:*

*Ls , ls -l and ls -la were applied and it is seen in especially /home and /tmp there were no strange files or directories Risk Level:*

*Zero For Now Recommendation:*

*Continue monitoring like regularly*

## **Finding 2: Permissions Investigation**

*Command Used:*

- i) `find /etc -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- ii) `find /tmp -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- iii) `find /home -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- iv) `find /var -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`

*Evidence:*

```
(kali㉿kali)-[~]
└─$ find /etc -type f -perm -002 2>>/dev/null

(kali㉿kali)-[~]
└─$ find /usr/bin -type f -perm -002 2>/dev/null

(kali㉿kali)-[~]
└─$ find /var -type f -perm -002 2>/dev/null

(kali㉿kali)-[~]
└─$ find /home -type f -perm -002 2>/dev/null
/home/kali/.insecure.txt

(kali㉿kali)-[~]
└─$ ls -la /home/kali/.insecure.txt
-rwxrwxrwx 1 kali kali 0 Apr 25 14:58 /home/kali/.insecure.txt
```

*It is seen from the observation that there a file placed in /home/kali named “insecure.txt” which have permissions for anyone to write so It is risky that a hacker can easily change and write his own script in this file which may be harmful / Risk Level:*

*Recommendation:*

### Finding 3: Suspicious Process

*ps aux | grep sleep Evidence:*

[illegible]

*Analysis:*

*As the observations show that there are two processes which are in sleep mode we can easily see their CPU usage, memory usage and also who created them when and how.*

*Risk Level:*

*Moderate*

*Recommendation:*

Deal with those processes if important let them finish if not kill them using commands :

- i) kill PID ii)
- kill -9 PID

**Finding 3: Failed Login Attempts**

*Command Used:*

- i) `grep "authentication failure" /var/log/auth.log` Evidence:

```

(kali㉿kali)-[~]
$ journalctl | grep -i auth | tail -20
Apr 24 08:20:05 kali su[35682]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=root
Apr 25 03:39:26 kali systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RP
CSEC_GSS skipped, unmet condition check ConditionPathExists=/etc/krb5.keytab
Apr 25 03:39:27 kali systemd[1]: Starting polkit.service - Authorization Manager ...
Apr 25 03:39:28 kali systemd[1]: Started polkit.service - Authorization Manager.
Apr 25 03:39:53 kali polkitd[445]: Registered Authentication Agent for unix-session:2 (sy
stem bus name :1.48 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/m
ate/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8)
Apr 25 04:06:12 kali su[20991]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=root
Apr 25 04:06:20 kali su[21045]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=root
Apr 25 04:06:30 kali su[21123]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=root
Apr 25 04:33:41 kali systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RP
CSEC_GSS skipped, unmet condition check ConditionPathExists=/etc/krb5.keytab
Apr 25 04:33:42 kali systemd[1]: Starting polkit.service - Authorization Manager ...
Apr 25 04:33:43 kali systemd[1]: Started polkit.service - Authorization Manager.
Apr 25 04:34:07 kali polkitd[463]: Registered Authentication Agent for unix-session:2 (sy
stem bus name :1.48 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/m
ate/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8)
Apr 25 04:55:09 kali xfce4-screensaver-dialog[12591]: pam_unix(xfce4-screensaver:auth): a
uthentication failure; logname=kali uid=1000 euid=1000 tty=:0.0 ruser= rhost= user=kali
Apr 25 14:40:03 kali systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RP
CSEC_GSS skipped, unmet condition check ConditionPathExists=/etc/krb5.keytab
Apr 25 14:40:05 kali systemd[1]: Starting polkit.service - Authorization Manager ...
Apr 25 14:40:06 kali systemd[1]: Started polkit.service - Authorization Manager.
Apr 25 14:40:40 kali polkitd[507]: Registered Authentication Agent for unix-session:2 (sy
stem bus name :1.47 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/m
ate/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8)
Apr 25 14:57:20 kali su[12715]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/1 ruser=kali rhost= user=root
Apr 25 14:57:25 kali su[12768]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/1 ruser=kali rhost= user=root
Apr 25 14:57:37 kali su[12828]: pam_unix(su:auth): authentication failure; logname=kali u
id=1000 euid=0 tty=/dev/pts/1 ruser=kali rhost= user=root

```

#### Analysis:

*The evidence shows that there are some authentication failure in our system we can see time , host , commands and even the name of the user who did it .*

#### Risk Level:

*High*

#### Recommendation:

*Immediately deal with it if unrecognized user is noticed then quickly remove that user form the system and observe the system with detail to see if he did some damage.*

#### Finding 4: Administrative Activity

#### Command Used:

- i) `Journalctl | grep -i auth.log`
- ii) `grep "sudo" /var/log/auth.log | sort | uniq -c | sort -nr | head` (For Counting Repeated Entries)
- iii) `tail -20 /var/log/auth.log`

(CHECKING THE RECENT LOGS)

#### Analysis:

As shown from the picture that we saw the logs produced while using sudo so we can see that who and when he/she tried or accessed admin .

#### Risk Level:

Moderate

#### Recommendation:

Continue Monitoring The recent lgos and admin logs to be safe always .

#### COMMANDS USED:

- 1) `ls` (List of all the visible files and directories )
- 2) `ls -l` (To See the permissions as well as owner)
- 3) `ls -la` (For hidden Directories and files )
- 4) `find /etc -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- 5) `find /tmp -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- 6) `find /home -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- 7) `find /var -type f -perm -002 2>/dev/null | grep -E "\.ssh | \.py | \.conf | \.service | \.txt"`
- 8) `ps aux`
- 9) `ps aux --sort=-%cpu | head` (Fot Seeing the top process for now )
- 10) `ps aux | grep sleep`

- 11) *grep "authentication failure" /var/log/auth.log*
- 12) *grep "sudo" /var/log/auth.log*
- 13) *grep "sudo" /var/log/auth.log | sort | uniq -c | sort -nr | head (For Counting Repeated Entries)*
- 14) *tail -20 /var/log/auth.log*
- 15) *Journalctl | grep -i auth.log*

### **FINAL CONCLUSION:**

So , From this project I learned and practice the file systems , some important commands of linux , saw and read logs , auth logs , admin logs , and learned how to read them how to observe the processes how to find weak permissions files their location and also learned how to write down the findings with professional approach .